Prediction of Progression-Free Survival in Breast Cancer Patients

William Donovan (wd2328), Hantang Qin (hq2229), Yongyan Liu (yl6107)

2026-03-23

Introduction

Progression-free survival (PFS) is a key clinical endpoint in breast cancer research, defined as the time from diagnosis to disease progression or death. In this report, we build a prediction model for log-transformed PFS using data from 2,000 breast cancer patients diagnosed between 1990 and 1995 (Part I), and evaluate whether this model generalizes to a separate cohort of 2,000 patients diagnosed between 2000 and 2005 (Part II).

We consider four candidate models that span a range of complexity: **elastic net** as a regularized linear baseline with built-in variable selection; **PLS** as a dimension-reduction approach suited to correlated predictors; **MARS** for automatically capturing nonlinearity and interactions while maintaining interpretable piecewise-linear structure; and **GAM** for flexible nonparametric smoothing of continuous predictors. This set covers the spectrum from linear to nonlinear methods, allowing us to assess whether additional model flexibility translates into meaningful predictive gains.

Part I: Prediction Model for PFS (1990–1995 Cohort)

Data Overview

The dataset (datA) contains 2,000 patients with complete follow-up and no missing values. The response variable is log-transformed PFS (months). There are 15 predictors: five continuous (age,

BMI, tumor size, number of positive lymph nodes, Ki-67 index) and ten categorical (menopausal status, race, tumor grade, ER/PR/HER2 status, clinical stage, surgery type, chemotherapy, and hormonal therapy). Table 1 summarizes the sample characteristics. Exploratory analysis revealed that log(PFS) is approximately normally distributed (Figure 1). Among continuous predictors, tumor size and lymph node count show negative associations with log(PFS), while correlations among predictors are generally low (Figures 2–3). Boxplots indicate that ER-positive status, lower tumor grade, and earlier clinical stage are associated with longer PFS (Figure 4).

Model Training

All analyses were conducted in R (version 4.5) using the `caret` package as the primary modeling framework, with `glmnet` for elastic net, `earth` for MARS, `mgcv` for GAM, and `pls` for PLS. Categorical predictors (menopause, race, tumor grade, ER, PR, HER2, stage, surgery, chemo, hormonal therapy) were converted to factors, and the ten resulting factor variables were expanded into dummy indicators via treatment coding (first level as reference). All models were evaluated using 10-fold CV with the same pre-specified fold indices (generated by `createFolds` with seed 8106) to ensure a fair comparison. The tuning parameter selection criterion was minimization of CV RMSE. After identifying the best hyperparameters, each final model was refit on the entire training set (datA, $n = 2000$).

Elastic net was tuned over a grid of mixing parameter $\alpha \in \{0, 0.1, \dots, 1\}$ (11 values from ridge to lasso) and regularization parameter λ over 100 values equally spaced on the log scale from e^{-6} to e^{2}. No additional centering or scaling was applied, as `glmnet` standardizes internally. The optimal model selected $\alpha = 0.7$ with $\lambda = 0.0159$, retaining 13 non-zero coefficients. For consistency and reproducibility, this coefficient count should be reported the same way throughout the report.

PLS was tuned over 1–15 components. Predictors were centered and scaled within each CV fold via `preProcess`. The best model used 12 components.

MARS was tuned over interaction degree $\in \{1, 2, 3\}$ and the maximum number of retained terms (nprune) $\in \{2, \dots, 18\}$, with internal generalized cross-validation (GCV) used for basis function pruning. The best model selected degree 1 (additive, no interactions) with nprune = 9, yielding 8

basis functions involving 6 predictors.

GAM was fitted via `mgcv::gam()` with thin-plate regression splines (`s()`) on the five continuous predictors (age, BMI, tumor size, lymph nodes, Ki-67) and parametric (linear) terms for all categorical predictors. Smoothing parameters were estimated by restricted maximum likelihood (REML). Since `caret`'s GAM wrapper applies smoothing to all columns including dummy variables, we instead performed manual 10-fold CV using the same fold indices to maintain proper handling of categorical terms.

Results

Cross-validated performance was similar across all four models, with only small differences in CV RMSE (Table 2, Figure 5). The small gap across models suggests that greater flexibility yields only modest predictive improvement in this cohort.

Final model selection Although the four candidate models had very similar CV RMSE values, MARS was selected as the final model because it achieved competitive predictive performance while offering better interpretability than GAM and more flexibility than strictly linear approaches such as elastic net and PLS. Since the performance differences were very small, we prioritized a model that could capture nonlinear effects while still being explained clearly in the report.

Model interpretation The elastic net coefficients identify the most influential predictors as ER status, HER2 status, tumor grade, hormonal therapy, tumor size, and lymph nodes. The MARS model supports the same overall pattern while allowing nonlinear effects for some continuous variables. In practical terms, patients with more positive lymph nodes and larger tumors tended to have shorter progression-free survival, while ER-positive patients tended to have better outcomes. HER2-positive disease and poor tumor differentiation were associated with worse prognosis in this earlier cohort. These findings are clinically reasonable because both tumor burden and aggressive tumor biology are closely related to cancer progression.

Variable importance rankings from MARS and elastic net also place lymph nodes, ER status, tumor grade, HER2 status, and tumor size among the strongest predictors (Figure 6). This gives a fairly coherent clinical picture across different model types, even though the exact mathematical

form differs. Partial dependence plots and smooth-function plots further suggest that some predictor effects are not strictly linear (Figures 7–8). That is one reason nonlinear methods remained competitive even when the overall performance gap was small.

Performance assessment The main performance metric reported for Part I should be CV RMSE, not a separate "test RMSE," because the Part I workflow is described as full-cohort 10-fold CV rather than an internal train/test split. If your current report still says "Test RMSE for the MARS model is 0.7504," that sentence should be deleted or replaced with a CV-based summary sentence. This is more consistent with the report logic and the rubric's request to report and assess model performance clearly.

Overall, the Part I results suggest that a small set of tumor and receptor-status variables captures an important part of PFS variation, while also showing that substantial unexplained variability remains, which is common in clinical prognosis settings.

Part II: Generalization to the 2000–2005 Cohort

Evaluating Model A on Data B

We applied all four models trained on datA to datB. As shown in Table 5, RMSE increased from approximately 0.70 (CV on datA) to approximately 0.84 on datB, representing a relative increase of roughly 20%. The R^2 dropped from approximately 0.20 to approximately 0.15. The observed-versus-predicted plot (Figure 10) reveals systematic underprediction for patients with longer PFS, and the residual distribution (Figure 11) is shifted positively (mean residual > 0), indicating that patients in the later cohort tend to have longer PFS than predicted by the 1990s model. These results demonstrate that the prediction model does not generalize well to the 2000–2005 cohort, which original model doesn't generalize well to datB.

Understanding the Distributional Shift

Comparing the two cohorts (Table 6, Figures 12–13), cohort B has a higher mean log(PFS) than cohort A, and several predictors also show shifted distributions. For example, cohort B has slightly lower average tumor size, fewer positive lymph nodes, lower Ki-67, and higher average BMI, while treatment patterns also changed across the two diagnosis periods. These differences are consistent with improved outcomes over time and suggest that the predictor-outcome relationships learned from the 1990–1995 cohort were not fully stable in the 2000–2005 cohort. This helps explain why the original model transported poorly to datB.

New Model for the 2000–2005 Cohort

We retrained all four models on datB using the same tuning grids and a new set of pre-specified 10-fold CV indices. As shown in Table 7, the candidate models again had very similar performance. PLS achieved the lowest CV RMSE (0.6649), while Elastic Net and GAM were essentially tied at 0.6659, and MARS remained close behind at 0.6680. Because the differences were very small, the overall conclusion is that no single method dominates strongly in the later cohort. If interpretability is prioritized, MARS remains a reasonable final reporting model because it still performs competitively while providing a clearer description of nonlinear effects and interactions. Key differences compared to Model A:

1. **Interaction effects**: MARS for cohort B selected interaction terms involving tumor size, lymph nodes, Ki-67, BMI, ER status, and hormonal therapy. This suggests that prognosis in the later cohort may depend more strongly on combinations of predictors rather than only additive effects.

2. **Hormonal therapy**: Its MARS coefficient increased from +0.137 to +0.425, and the elastic net coefficient for hormonal therapy also increased from +0.0844 in cohort A to +0.1550 in cohort B. This may reflect a stronger prognostic role of hormonal therapy in the later cohort.

3. **HER2 status**: HER2 was a negative predictor in cohort A but became much less important in cohort B. In the elastic net comparison, the coefficient changed from −0.1222 to −0.0006,

and HER2 was absent from the final MARS description for cohort B. This pattern may be related to changes in HER2-targeted treatment during the later period.

4. **Variable importance shift**: In cohort A, lymph nodes was among the strongest predictors, while the later-cohort model emphasized a somewhat different predictor profile. This indicates that the relative prognostic importance of clinical variables shifted across cohorts(Figure 15).

5. **Broader predictor set**: Elastic net for cohort B retained 15 non-zero coefficients (vs. 11), newly including menopause, PR status, and clinical stage.This is consistent with a somewhat broader and more complex prediction structure in the later cohort.

In conclusion, the 1990–1995 prediction model does not generalize well to the 2000–2005 cohort. Performance worsened on datB, the residuals show systematic underprediction, and the later cohort differs both in outcome distribution and in the apparent importance of several predictors. The stronger role of hormonal therapy, the reduced importance of HER2, and the emergence of interaction effects suggest that treatment patterns and predictor-outcome relationships changed over time. A cohort-specific model is therefore more appropriate for the later population. More broadly, this result shows that clinical prediction models may lose accuracy when applied to future cohorts when prognosis and treatment practice evolve.

Appendix: Figures and Tables

Tables

Table 1: Sample characteristics (datA, n = 2000). Continuous variables reported as Mean (SD); categorical variables as Level: n (%).

Variable	Summary
Continuous	
Ki67	21.85 (11.58)
age	54.99 (11.65)
bmi	26.06 (4.56)
log_PFS	2.2 (0.78)
lymph_nodes	2.62 (2.59)
tumor_size	3.38 (1.71)
Categorical	
ER	0: 719 (36%); 1: 1281 (64%)
HER2	0: 1556 (77.8%); 1: 444 (22.2%)
PR	0: 1039 (52%); 1: 961 (48%)
chemo	0: 895 (44.8%); 1: 1105 (55.2%)
hormonal_therapy	0: 1066 (53.3%); 1: 934 (46.7%)
menopause	0: 689 (34.5%); 1: 1311 (65.6%)
race	1: 1284 (64.2%); 2: 289 (14.4%); 3: 176 (8.8%); 4: 251 (12.6%)
stage	1: 504 (25.2%); 2: 893 (44.6%); 3: 603 (30.1%)
surgery	0: 986 (49.3%); 1: 1014 (50.7%)
tumor_grade	1: 383 (19.1%); 2: 871 (43.5%); 3: 746 (37.3%)

Table 2: Cross-validated performance of candidate models (datA).

Model	CV_RMSE	CV_R2
ElasticNet	0.6906	0.2016
PLS	0.6917	0.1989
MARS	0.6916	0.1967
GAM	0.6872	0.2064

Table 3: Non-zero Elastic Net coefficients (datA).

Variable	Coefficient
(Intercept)	2.6319
ER1	0.2389
HER21	-0.1222
tumor_grade3	-0.1092
hormonal_therapy1	0.0844
lymph_nodes	-0.0563
tumor_size	-0.0562
race4	-0.0414
race2	0.0238
surgery1	-0.0186
race3	0.0163
menopause1	-0.0110
Ki67	-0.0036
age	-0.0021

Table 4: MARS basis function coefficients (datA).

Term	Coefficient
(Intercept)	2.1124
h(8-lymph_nodes)	0.0709
ER1	0.3336
h(bmi-23.9)	-0.0228
h(23.9-bmi)	-0.0442
HER21	-0.1642
tumor_grade3	-0.1589
h(tumor_size-0.4)	-0.1771

Term	Coefficient
h(tumor_size-1.9)	0.1421

Table 5: Generalization performance of Model A on datB.

Model	CV_RMSE_A	RMSE_on_B	R2_on_B	Pct_Increase
ElasticNet	0.6906	0.8305	0.1529	20.3%
PLS	0.6917	0.8340	0.1478	20.6%
MARS	0.6916	0.8469	0.1471	22.5%
GAM	0.6872	0.8461	0.1448	23.1%

Table 6: Mean (SD) comparison of continuous variables between cohorts.

Variable	Cohort A (1990-1995)	Cohort B (2000-2005)
age	54.99 (11.65)	56.05 (11.22)
bmi	26.06 (4.56)	27.44 (5.05)
tumor_size	3.38 (1.71)	3.08 (1.68)
lymph_nodes	2.62 (2.59)	2.32 (2.53)
Ki67	21.85 (11.58)	20.37 (11.71)
log_PFS	2.2 (0.78)	2.74 (0.74)

Table 7: Cross-validated performance of models on datB.

Model	CV_RMSE	CV_R2
ElasticNet	0.6659	0.1715
PLS	0.6649	0.1737
MARS	0.6680	0.1675

Model	CV_RMSE	CV_R2
GAM	0.6659	0.1727

Table 8: Elastic net coefficient comparison between cohorts.

Variable	Cohort_A	Cohort_B	Change
(Intercept)	2.6319	3.0746	0.4427
HER21	-0.1222	-0.0006	0.1216
ER1	0.2389	0.1350	-0.1039
race4	-0.0414	0.0511	0.0925
stage3	0.0000	-0.0786	-0.0786
PR1	0.0000	0.0768	0.0768
hormonal_therapy1	0.0844	0.1550	0.0706
tumor_grade2	0.0000	0.0609	0.0609
race2	0.0238	-0.0278	-0.0516
race3	0.0163	-0.0338	-0.0501
lymph_nodes	-0.0563	-0.0240	0.0323
menopause1	-0.0110	-0.0299	-0.0189
tumor_size	-0.0562	-0.0403	0.0159
stage2	0.0000	0.0121	0.0121
chemo1	0.0000	-0.0059	-0.0059
tumor_grade3	-0.1092	-0.1048	0.0044
bmi	0.0000	-0.0035	-0.0035
surgery1	-0.0186	-0.0211	-0.0025
Ki67	-0.0036	-0.0042	-0.0006
age	-0.0021	-0.0024	-0.0003

Figures

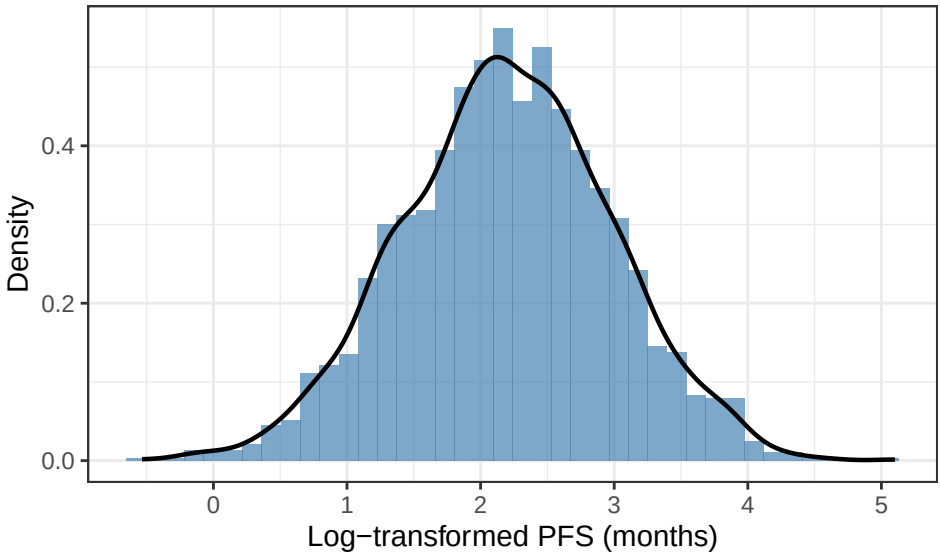

Figure 1: Distribution of log-transformed PFS (datA).

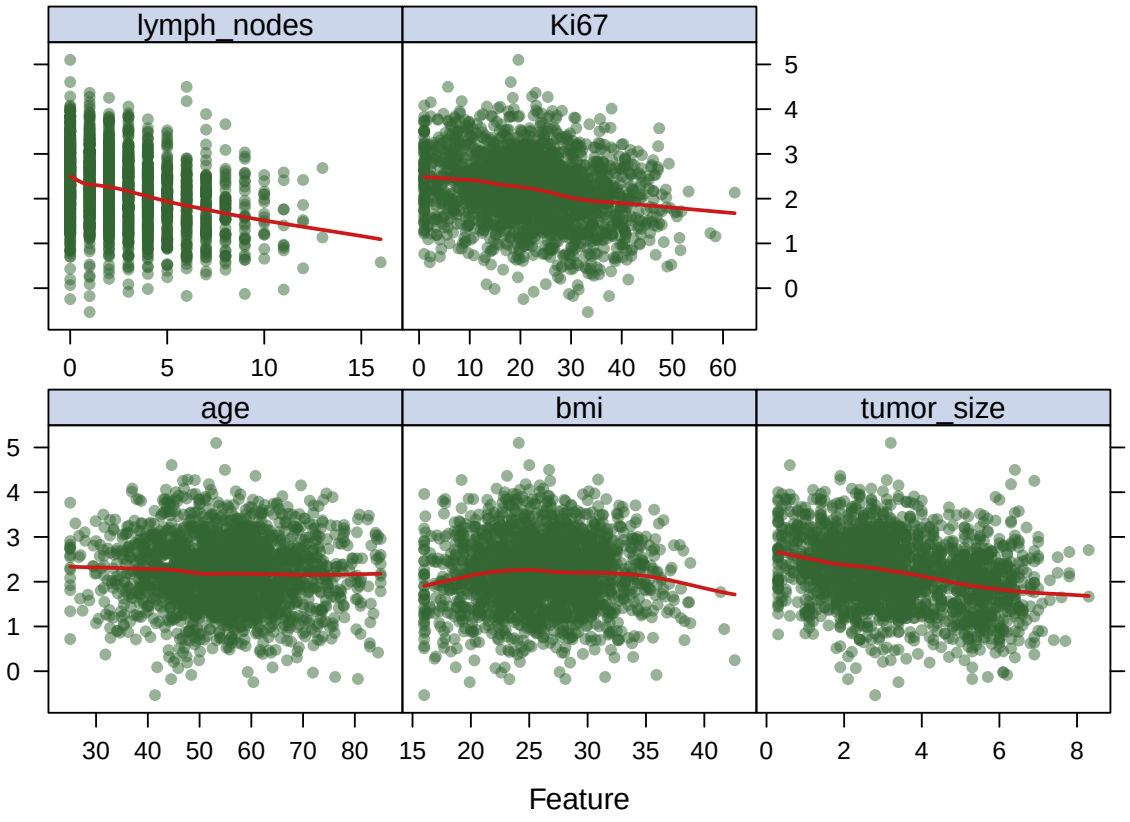

Figure 2: Scatter plots of continuous predictors vs. log_PFS.

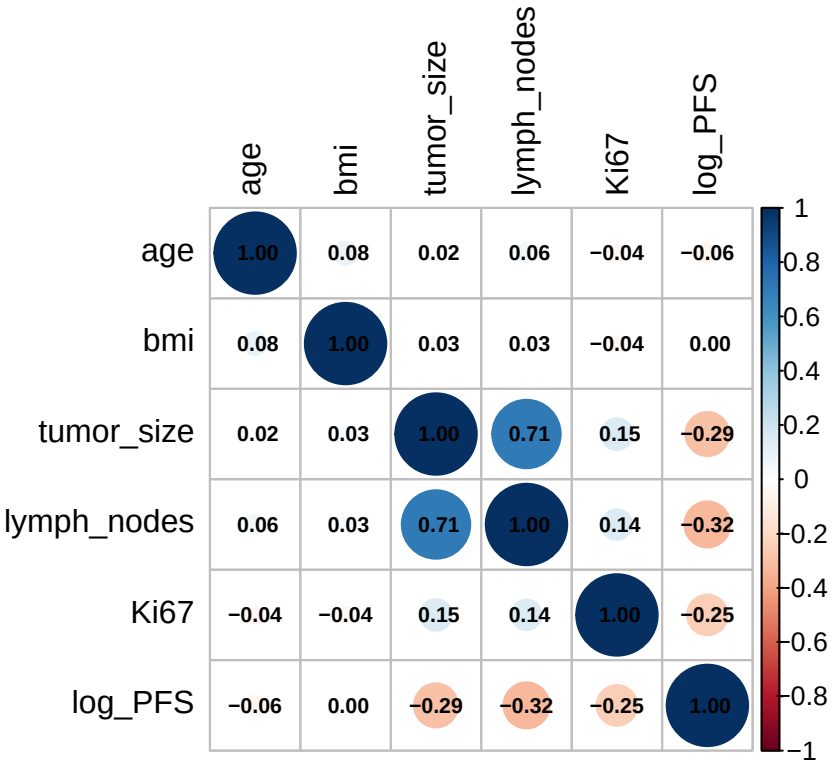

Figure 3: Correlation matrix of continuous variables.

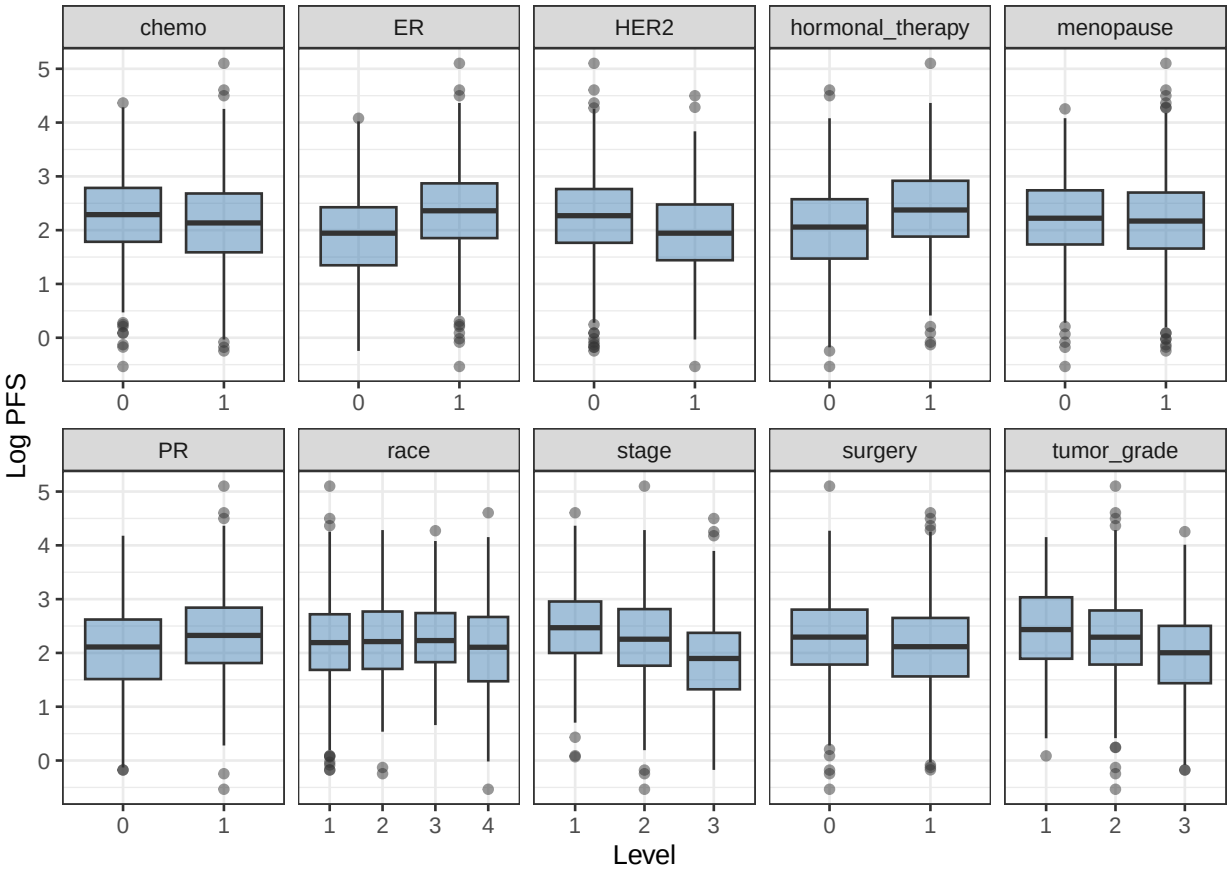

Figure 4: Boxplots of log_PFS by categorical predictors.

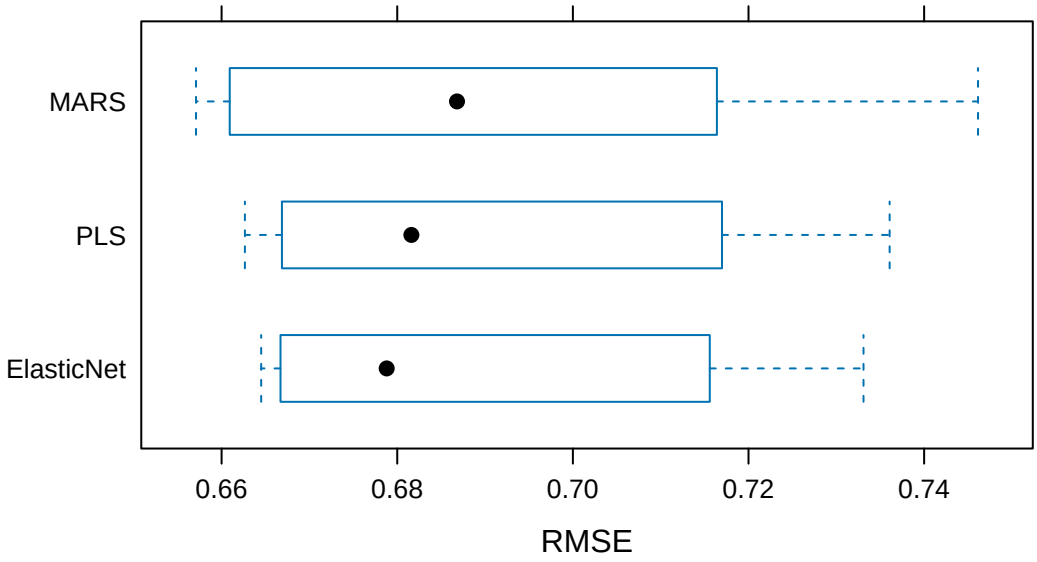

Figure 5: Cross-validated RMSE comparison (datA).

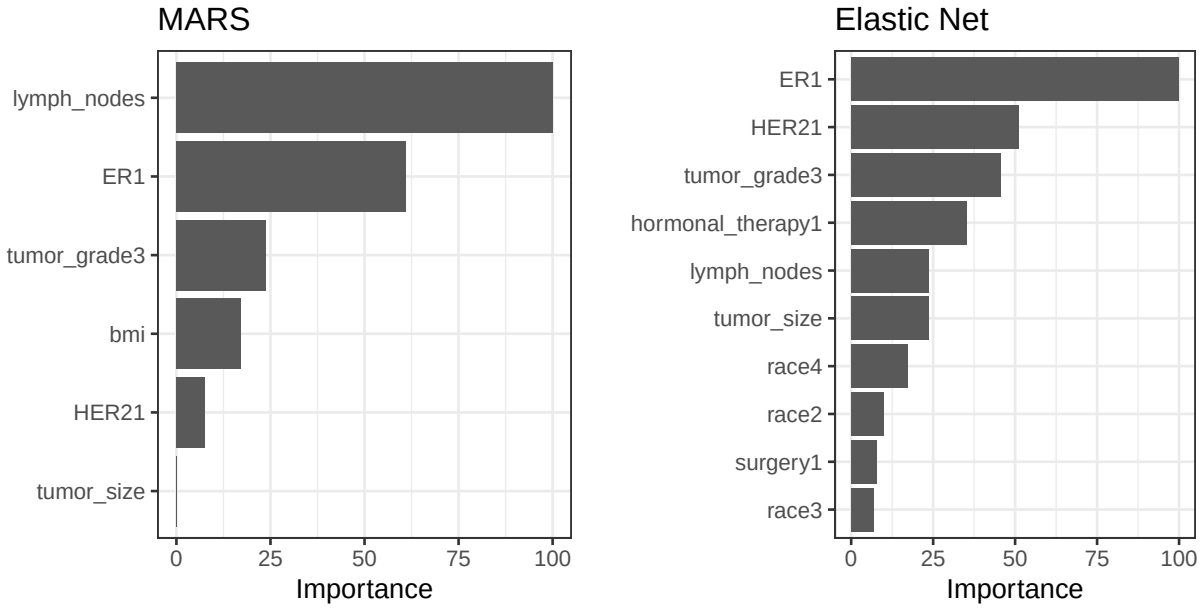

Figure 6: Variable importance from MARS and Elastic Net (datA).

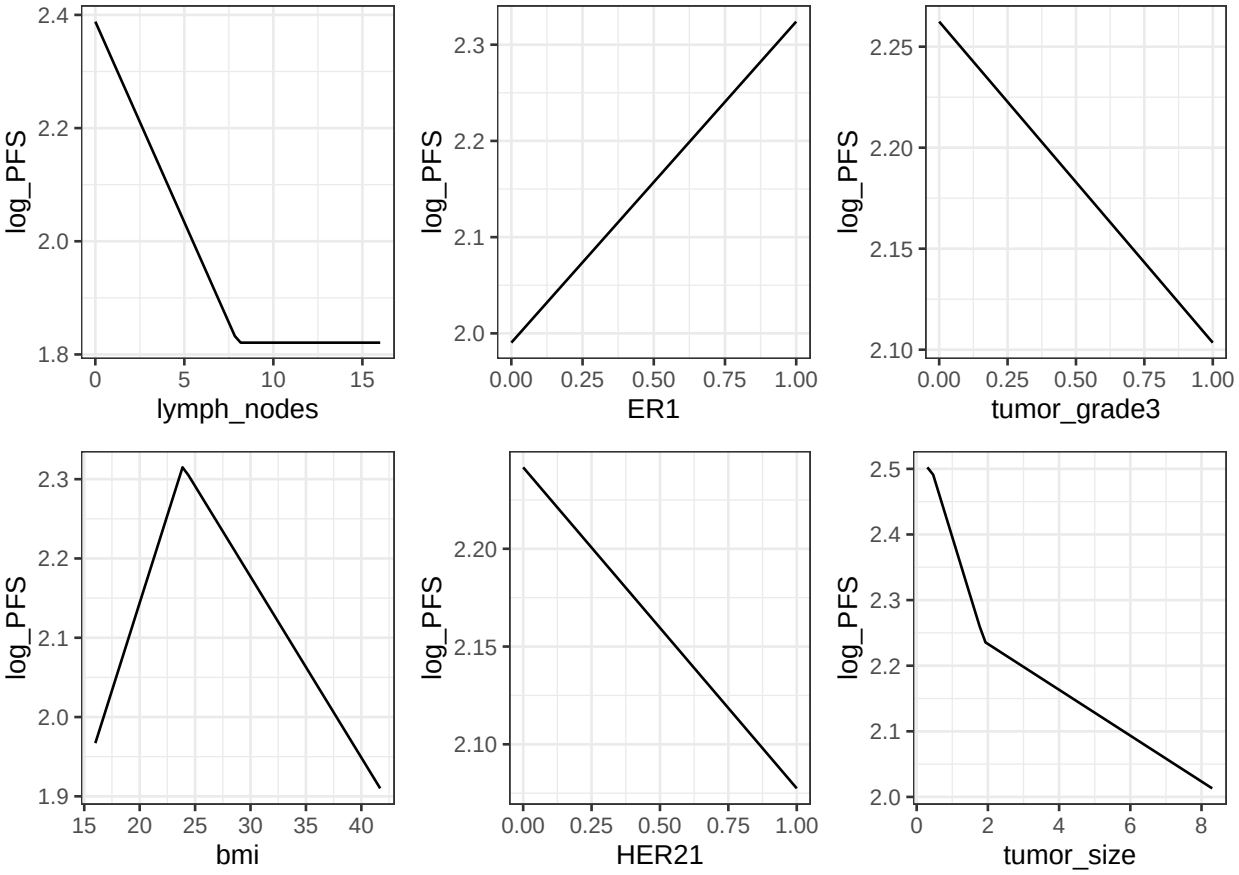

Figure 7: Partial dependence plots for top 6 predictors (MARS, datA).

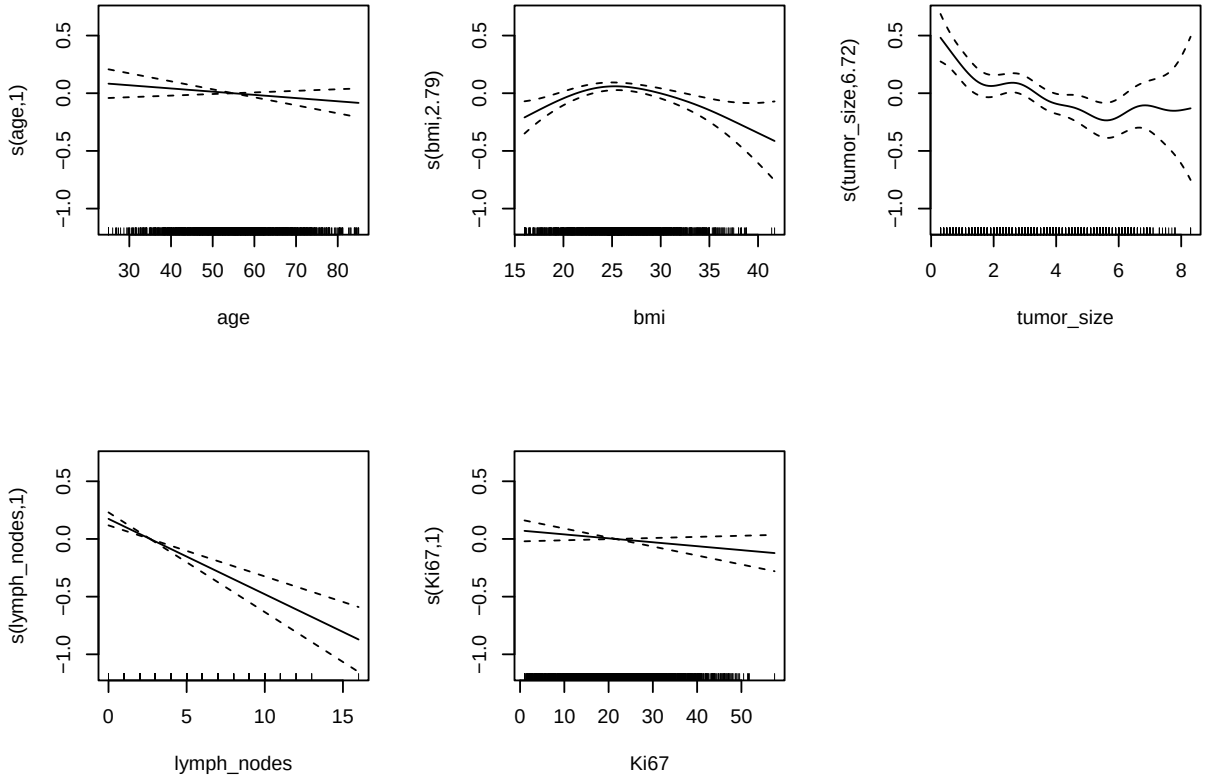

Figure 8: GAM smooth function estimates (datA).

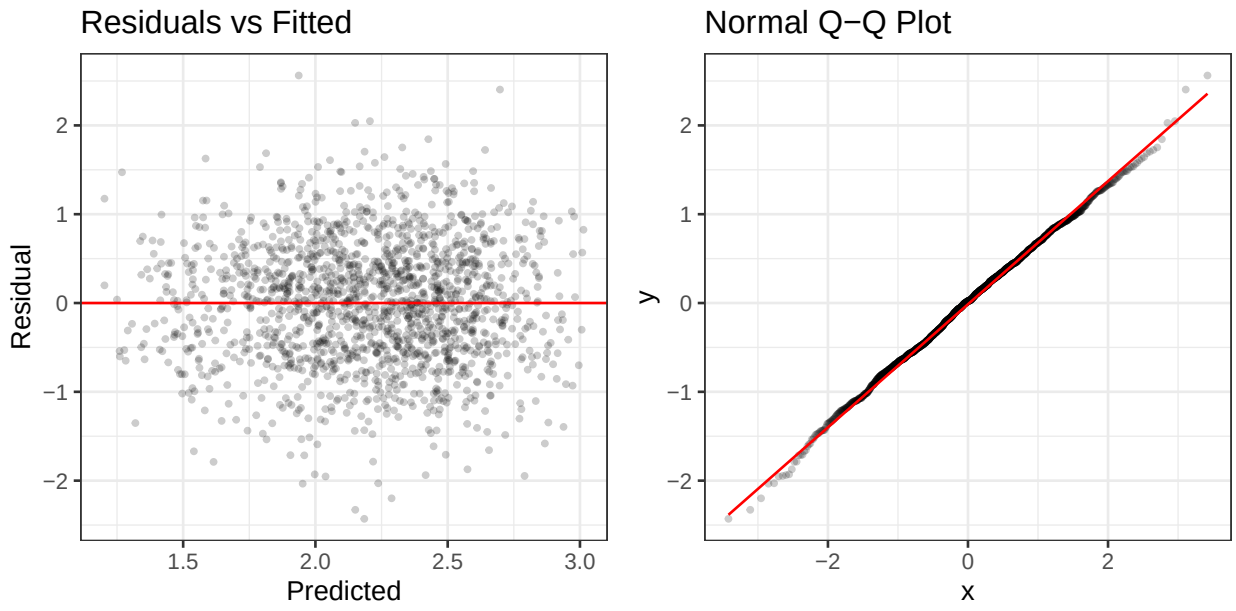

Figure 9: Residual diagnostics for MARS on training data.

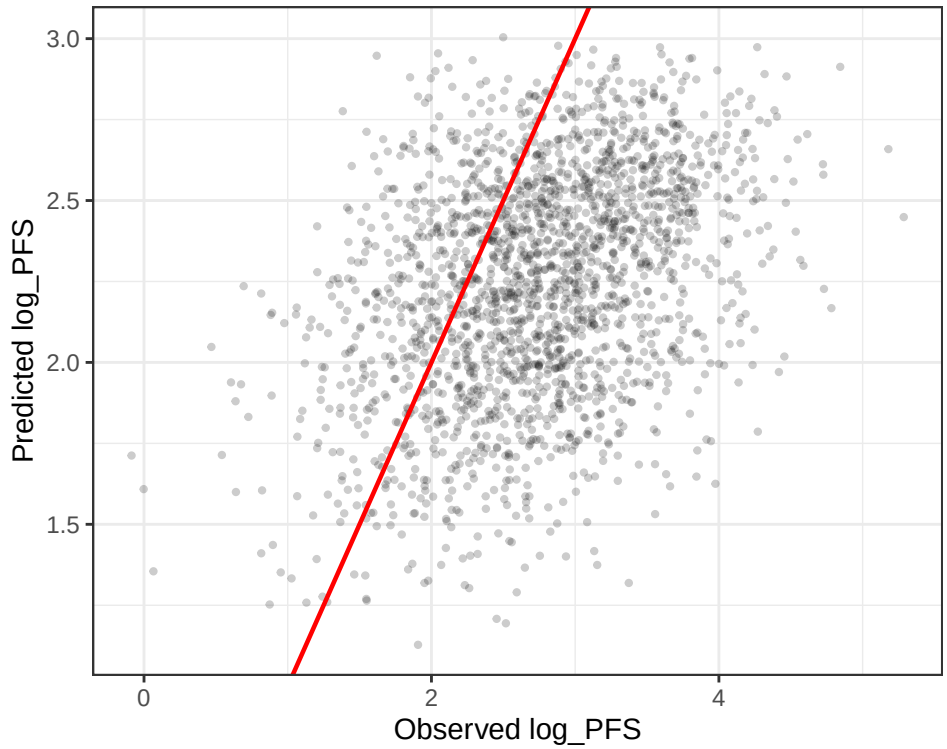

Figure 10: Observed vs. predicted log_PFS on datB (MARS from cohort A).

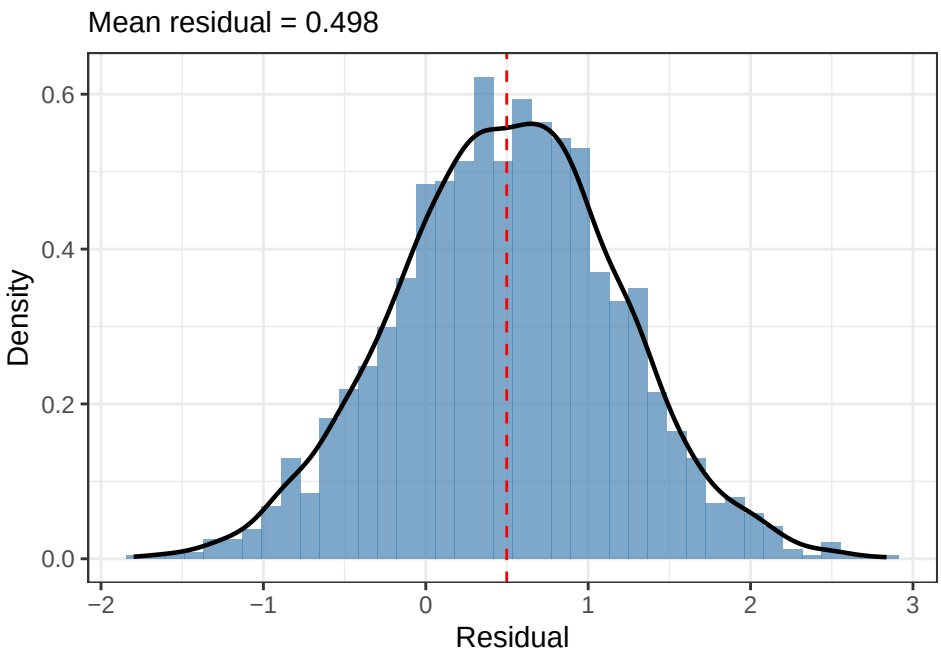

Figure 11: Residual distribution of Model A applied to datB.

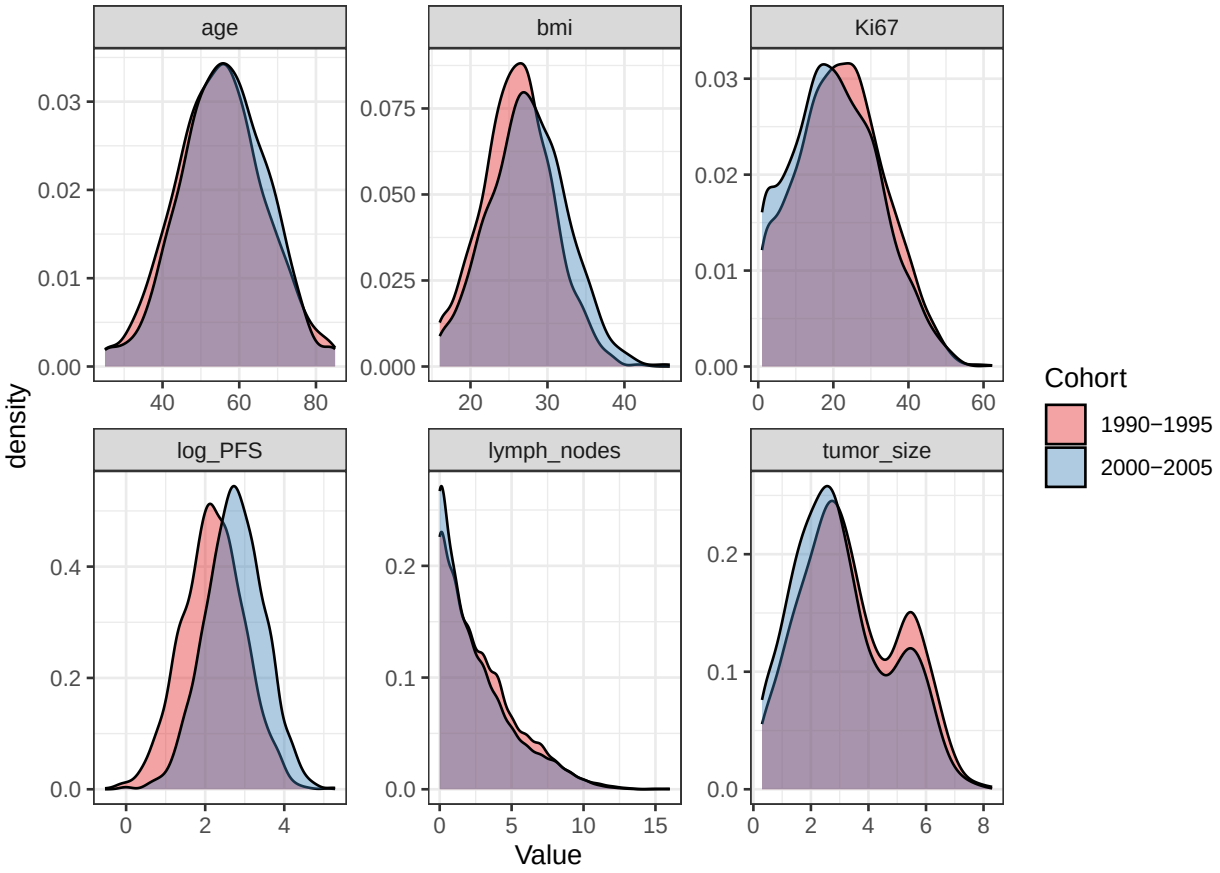

Figure 12: Distribution comparison of continuous variables between cohorts.

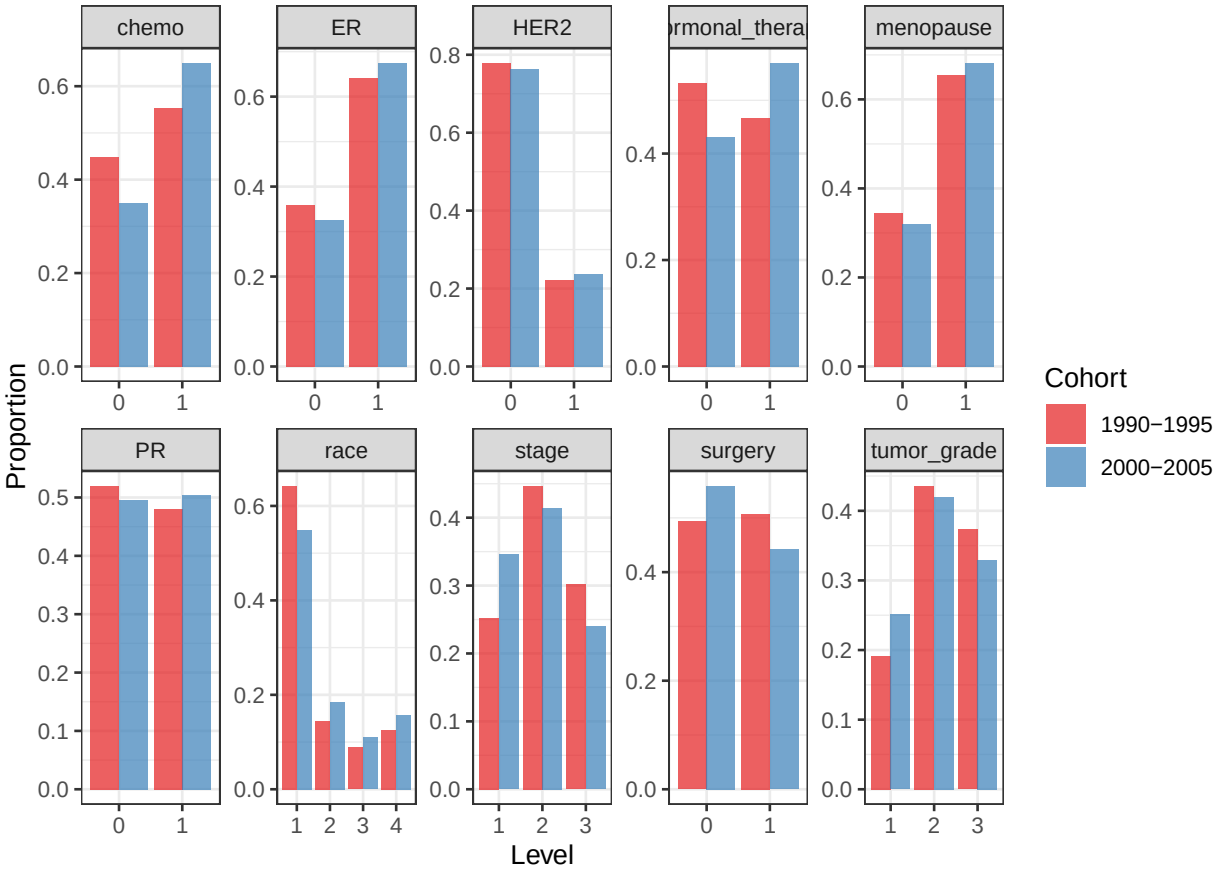

Figure 13: Categorical variable distributions across cohorts.

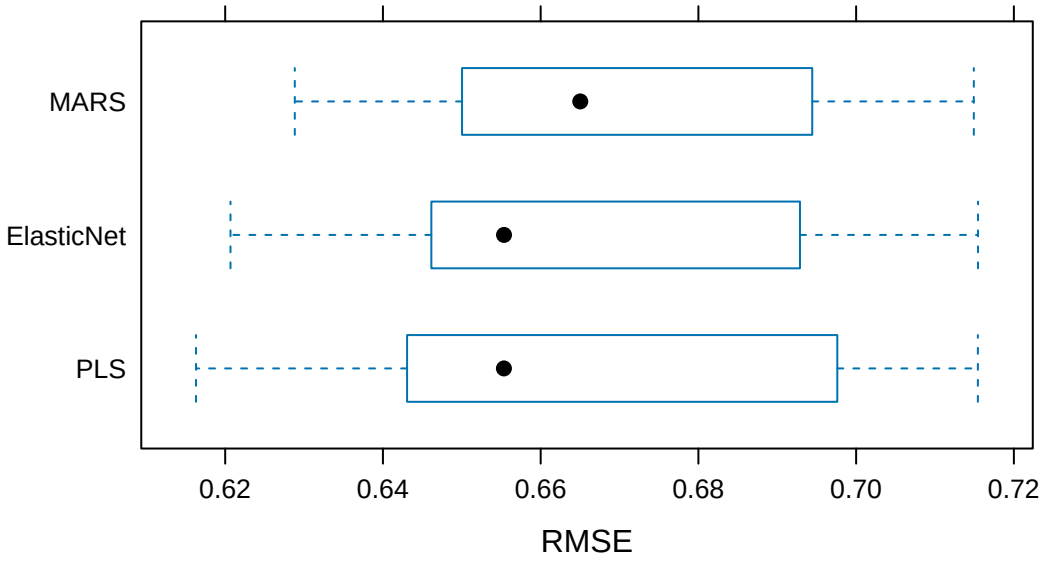

Figure 14: CV RMSE comparison for models trained on datB.

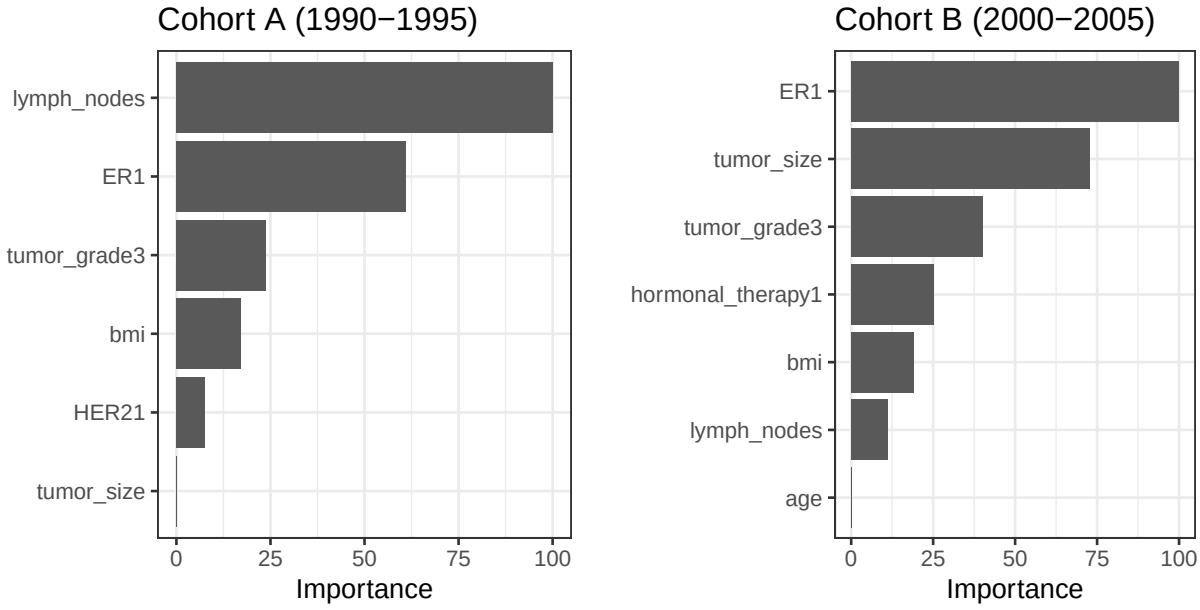

Figure 15: Variable importance comparison (MARS) between cohorts.